

Nuclear Fission and Chain Reactions

NE 630

Lesson 02

Prepare: Read FNRP Sections 1.4–1.5. Revisit binding energy per nucleon from Lesson 1 and define the key terms before class.

Macro objective

Characterize the neutron population and energy produced as functions of time using the neutron lifetime, the energy released per fission, and the multiplication factor k .

Vocabulary: fusion; fission; fission product; fission-product yield; prompt; delayed; decay heat; fissionable; fissile; fertile; generation; neutron lifetime; multiplication factor; criticality

1 Stability, fusion, and fission

Binding-energy relations

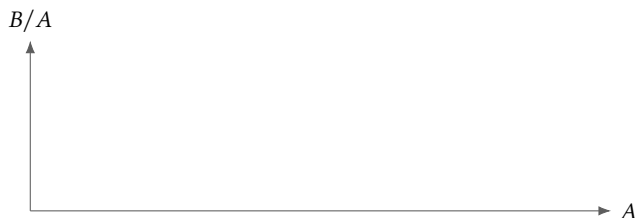
Using *nuclear masses*,

$$B = [Zm_p + Nm_n - m_X]c^2.$$

Using the *atomic mass* $m({}_Z^AX)$ and neglecting electron binding energies,

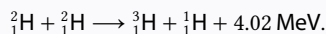
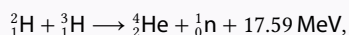
$$B = [Zm({}_1^1\text{H}) + Nm_n - m({}_Z^AX)]c^2, \quad \frac{B}{A} = \frac{B}{N+Z}.$$

Read the binding-energy curve



Why can either process release energy? What competing effects produce the maximum near the middle of the chart?

Two fusion examples from the notes



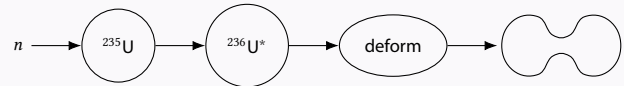
The energy is favorable; bringing two positive nuclei close enough to react is the engineering challenge. For D–T fusion, estimate the Coulomb energy $V_C = (4\pi\epsilon_0)^{-1}e^2/r$ at $r \approx 4 \text{ fm}$ and convert it to an equivalent temperature $T = V_C/k_B$; what does the resulting $\approx 0.36 \text{ MeV}$ ($\approx 4 \times 10^9 \text{ K}$) tell you about the need for quantum tunneling?

Three forms of nuclear power

1.
2.
3.

2 Anatomy of neutron-induced fission

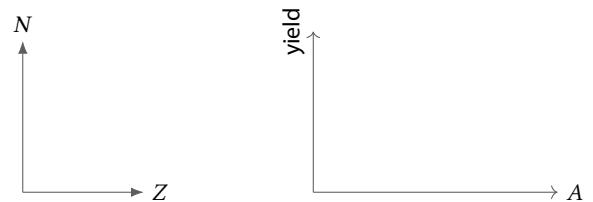
Prompt sequence and time scales



A heavy nucleus usually splits into two energetic fragments. Excitation energy produces prompt γ rays and approximately neutrons (about on average in the board sketch).

The split occurs within roughly ; prompt emissions are complete within roughly of the initiating neutron absorption.

Why are fission products neutron rich?



The parent has a large neutron-to-proton ratio. Splitting it near the usual two fission-yield peaks leaves fragments on which side of the stability line? How do they move toward stability?

Prompt versus delayed

Sort the emissions or energy carriers discussed on the board:
prompt fragment K.E.;

delayed β ; γ ; delayed n ;

Which contribution is effectively lost for heat production?

.....

Reading completion: fuel vocabulary

Fissionable:

Examples:

Fissile:

Examples:

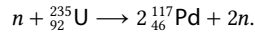
Fertile:

Pathways:

3 Energy release and decay heat

Symmetric-fission bookkeeping

Consider



$m({}^{235}\text{U})$
 $m({}^{117}\text{Pd})$
 m_n

235.043928 u
 116.917956 u
 1.0086649 u

$$Q_{\text{prompt}} = [m({}^{235}\text{U}) - 2m({}^{117}\text{Pd}) - m_n]c^2,$$

$$= \dots\dots\dots$$

Show the mass cancellation and arithmetic here:

After the neutron-rich fragments decay

For each fragment,



With $m({}^{117}\text{Sn}) = 116.902955 \text{ u}$,

$$Q_{\text{total}} = [m({}^{235}\text{U}) - 2m({}^{117}\text{Sn}) - m_n]c^2,$$

$$= \dots\dots\dots$$

Why can atomic masses be used directly for the complete β^- chain?

Fission-energy scale

The board estimate is roughly released per thermal fission of ${}^{235}\text{U}$, with about recoverable. Most prompt energy appears as fragment kinetic energy; each prompt neutron carries roughly

Decay heat immediately after shutdown

Delayed β and γ energy continues as fission products and their daughters decay. The board note marks an immediate decay-heat level of about of the pre-shutdown power:

$$P_{\text{decay}}(0^+) \approx \dots\dots\dots P_0.$$

For a reactor operating at P_0 , what heat-removal obligation remains after fission power is terminated?

4 Multiplication and chain reactions

Generation model

$$k = \frac{n_i}{n_{i-1}}, \quad n_i = n_0 k^i.$$

If one generation lasts ℓ seconds, then $i = t/\ell$ and

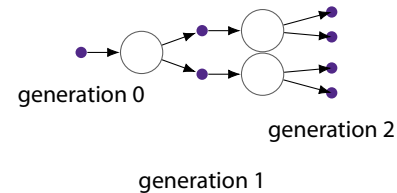
$$n(t) = n_0 k^{t/\ell} = n_0 \exp\left(\frac{\ln k}{\ell} t\right),$$

$$n(t) \approx n_0 \exp\left(\frac{k-1}{\ell} t\right) \quad (k \approx 1),$$

$$\frac{dn}{dt} \approx \frac{k-1}{\ell} n(t).$$

Here k is the generation-to-generation multiplication factor and ℓ is the neutron lifetime used in this simple model.

Build the generation picture



Which neutrons count toward k ? Why is "neutrons emitted per fission" not itself the reactor multiplication factor?

Criticality table

condition	name	neutron population
$k < 1$		
$k = 1$		
$k > 1$		

Connect neutron population to power

$$P(t) = \dots\dots\dots,$$

$$\frac{P(t)}{P(0)} \approx \dots\dots\dots.$$

Why can even a small departure of k from one matter when ℓ is short?

The three controls

E_f sets energy per fission; k sets the generation-to-generation change; ℓ sets the time scale. Together they set the scale and rate of energy production.